IERG4320/IEMS5726: Data Science in Practice

Lecture 13: Course Summary

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Only Practical Final Examination

Read the rules of final exam.

Except if you have medical or technical issues, you need to apply to Danny. Quite slow to arrange make-up, or you may have the make-up examination at night, or holiday:

http://www.res.cuhk.edu.hk/en-gb/examinations

Make-up paper will share the same format with the original examination paper.

Examination Schedule

- Date: 15th April, 2026 (Wednesday)
- Time: Session A1: 6:00 - 8:00 pm, Session A2: 8:30 - 10:30 pm
- Location: Room 1004, William M. W. Engineering Building (ERB 1004)

- Please arrive ERB 1004 around 20 minutes earlier to set up your computer.

Examination Format

- There will be 3 types of problems

1. Python Programming Problems (3 problems) - 30%
    - Data Preprocessing, Machine Learning, Data Visualization
    - Fill in the blanks
2. Short Questions - 20%
    - Many small problems not covered by the scenario-based problems
3. Scenario-based Written Problems (4 problems) - 50%
    - Around beginner to moderate interview questions

If you want an immediate FAIL from the course

- You cheat in the examination
- You communicate with any other people during the examination
- You answer phone calls or use your phone during the examination

Reminder: Other requirements for this course

- You will fail this course if …

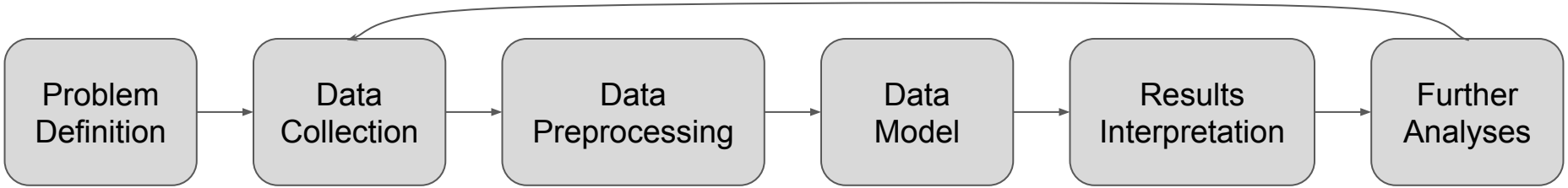
1. Your final examination performance is unsatisfactory (e.g. below 30%)
2. You did not submit the course project
3. Your grand total cannot meet the passing line (e.g. below 40%)

Some Tips

1. Revise the question 1 in the assignments
2. Revise examples in the lecture notes
3. Revise UReply questions (You need to make your own notes)
4. Search and prepare some interview questions (beginner and moderate level) related to data science
5. Understand the design of the course project

Course Summary

Basic Data Science Pipeline

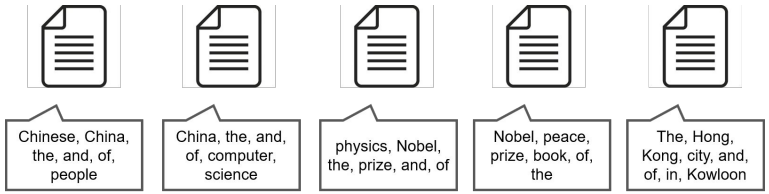


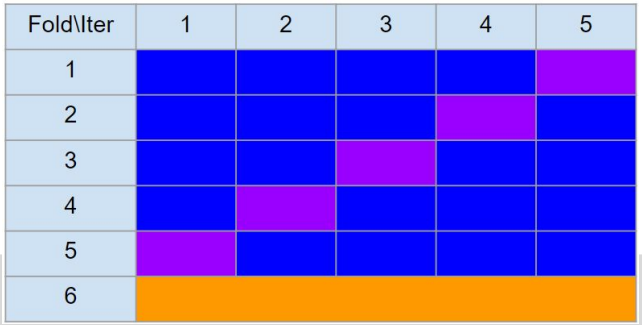
- For each of the case studies, can you identify the each of the above component?
- What are the necessary domain specific knowledge of the case studies?
- What are the four different types of data analytics?

Data Preprocessing

1. Load the libraries and dataset
2. Perform data wrangling
3. Check out the missing values
4. Handle the categorical values
5. Handle natural language, images and audio signals
6. Perform dimensionality reduction, if needed
7. Split the dataset into training, validation and testing set
8. Perform feature scaling

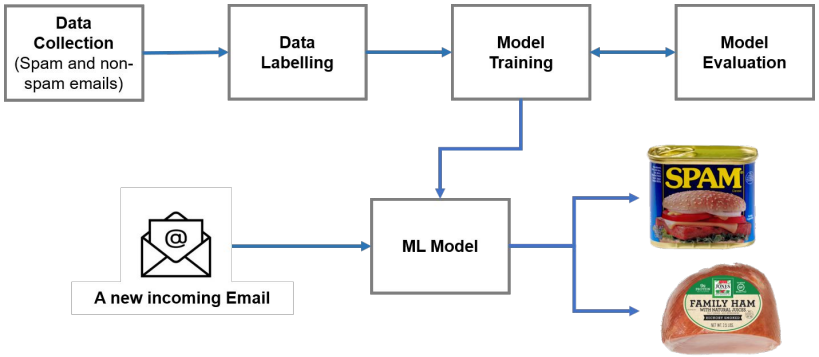
Name	Phone	Birth Date	State
John, Smith	445-881-4478	August 12, 1989	Maine
Jennifer Tal	+1-189-456-4513	11/12/1965	Tx
Gates, Bill	(876)546-8165	June 15, 72	Kansas
Alan Fitch	5493156648	2-6-1985	Oh
Jacob Alan	156-4896	January 3	Alabama

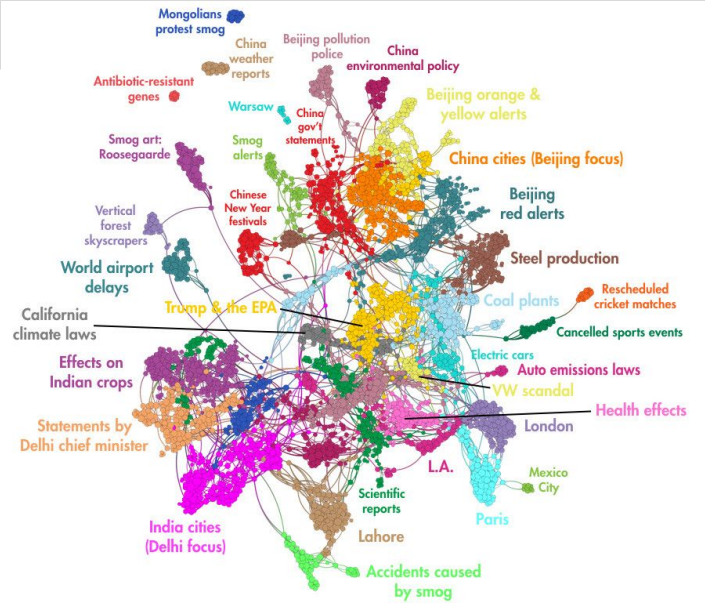


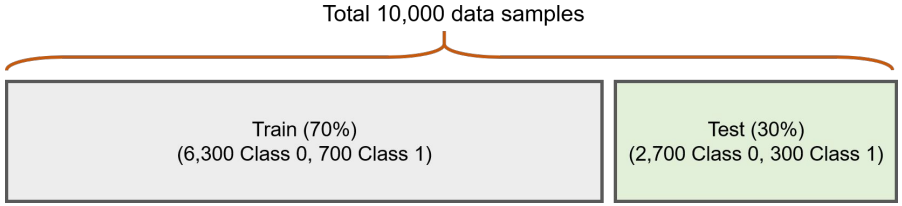


Basic Machine Learning

- 4 different types of data format
- 4 different types of machine learning problems
- 3 different types of machine learning algorithms
- Some common machine learning algorithms
- Different evaluations and metrics
- 3 different types of datasets

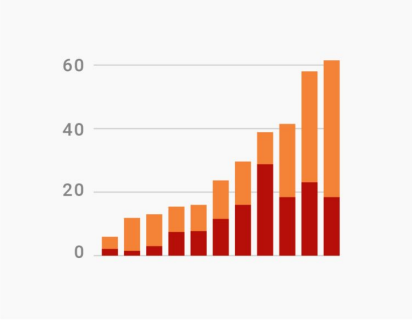






Data Visualization

- Visualization at different stages in data science
- Different types of plots / graphs for different purposes
- Some advices for good visualizations



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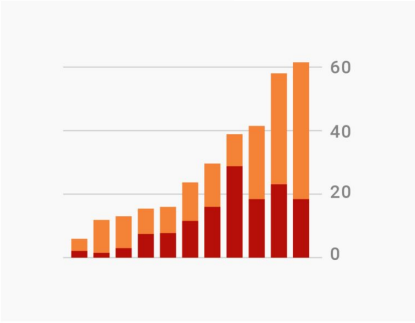

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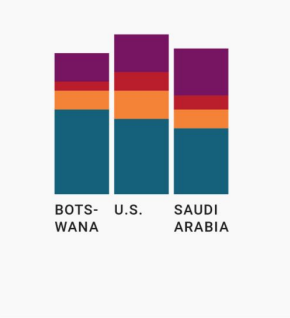

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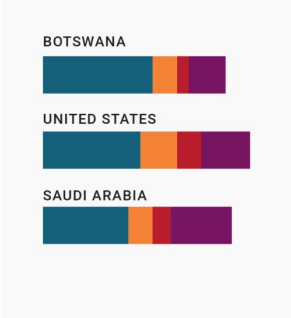
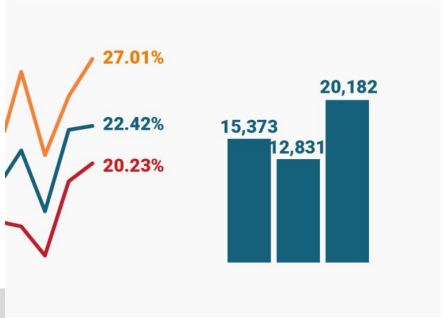


EVEN BETTER



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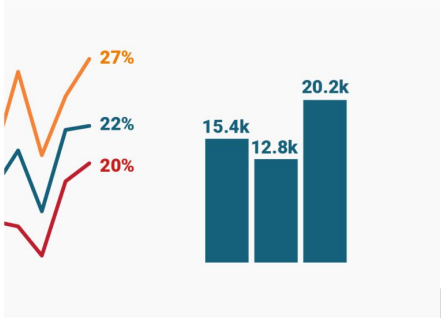


BETTER

Approach of this course: Case Studies

Q: What have we learnt about deep learning in this course?

1. Multilayer perceptron (**MLP**) and Deep Q-learning (**DQN**) agent in basic machine learning examples
2. Deep Cooperative Neural Networks (**DeepCoNN**), **Wide and Deep Network** and Graph convolutional network (**GCN**) in recommender system
3. Recurrent neural network (**RNN**), long short-term memory (**LSTM**) model and Encoder-based Transformers (**BERT**) in semantic analysis on NLP
4. Generated Pre-Trained Transformer (**GPT**) in text generation
5. Tacotron 2.0 in speech synthesis and HuBERT in speech recognition
6. Convolutional neural network (**CNN**) in image captioning
7. Generative adversarial network (**GAN**) and **Diffusion Model** in image generation
8. **D3QN**, **A3C** and **AlphaZero** agents in the game applications
9. **TripoSR** and **Shap-E** in 3D model generation

- This course introduces **17 deep learning models** using real examples.

(My Opinion) 3 Levels of Understanding the DL Models

1. **High-level understand**: Anyone can self-study this part
   - How can we use it?
   - What are the inputs and outputs?
2. **Intermediate level understand**: Approach of this course
   - How/Why does it work?
   - What are the mathematical models behind?
   - What are the parameters and hyperparameters?
3. **Complete/throughout understand**: Approach of most UG/PG courses
   - How can the formulae be updated?
   - Can we implement the model from scratch?

Future of Data Science

Data Science is just the Starting Point

- Some example outputs from data science or data analytics:
  - Business forecasts using some machine learning algorithms
  - Recommender systems
  - Anti-fraud/abuse models
  - Data mining/deep learning-extracting information from user submitted content, classifying images,etc
- These are programs, these are deliverables, but they are not products:
  - Web / mobile / console applications
  - Robots
  - Surveillance systems
  - … etc

Data Science, ML Engineer, Data Engineer

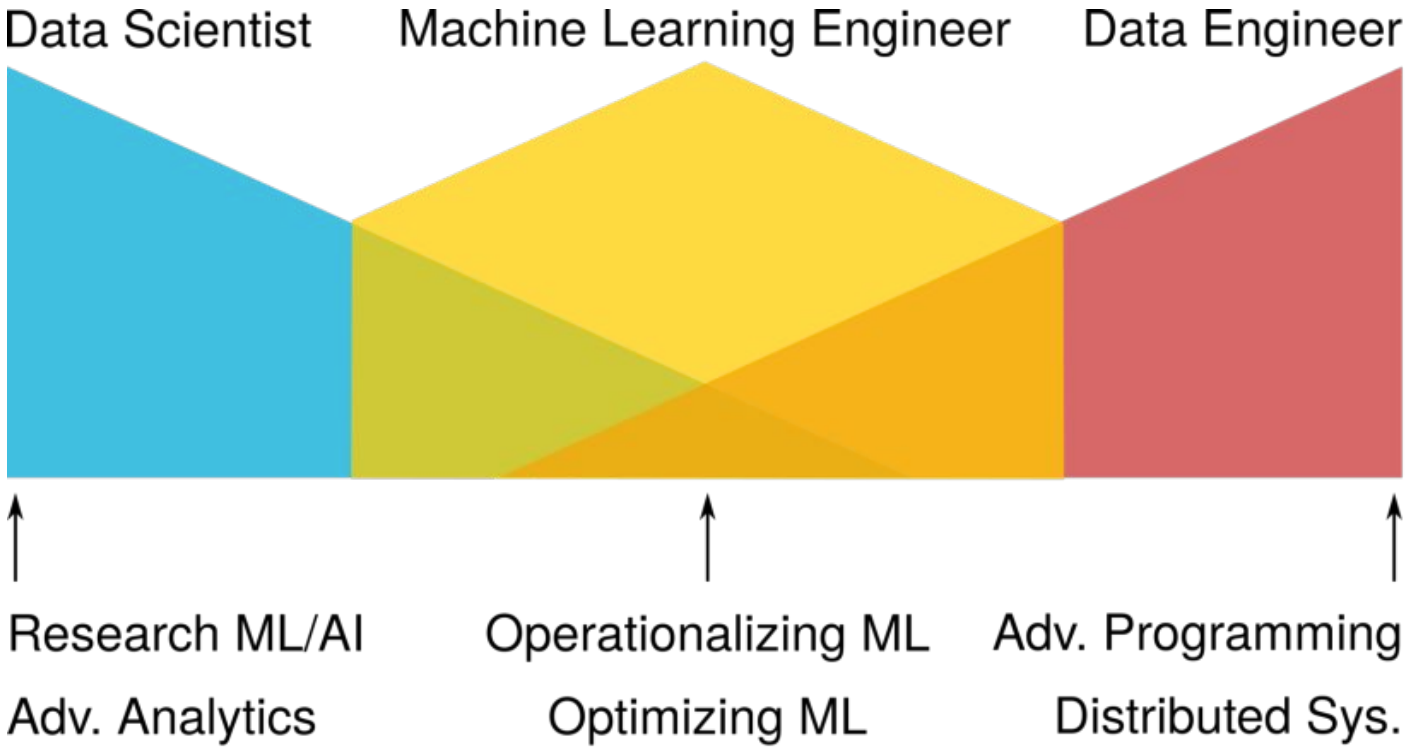


Image credit: https://medium.com/data-engineer-things/introduction-to-data-engineering-e16c9942dc2c

Example in Data Engineer: Fast Retailing

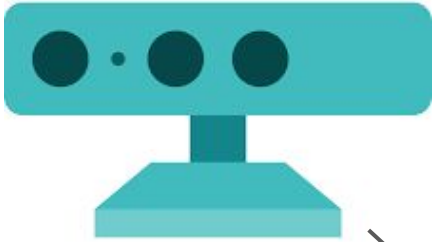
Clothes with **RFID**

Distributed system

Manufacturers can estimate what types of clothes to produce

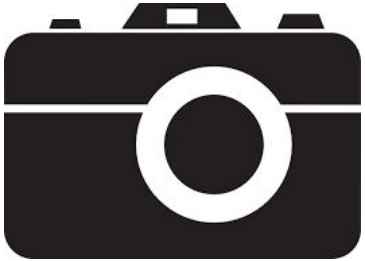
Shopkeepers can monitor the stock of the shop

Example of Internet-of-Thing: Smart Applications

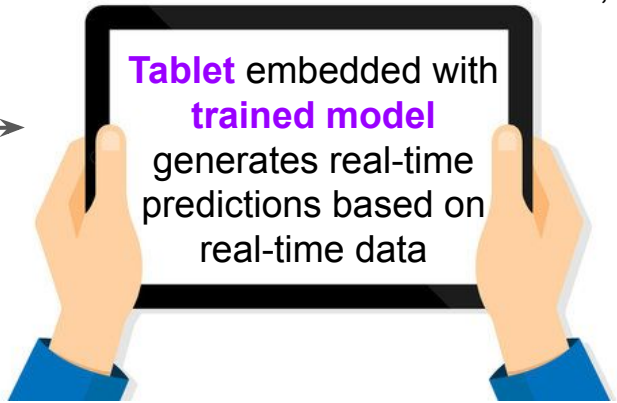


Digital sensor collects O_2/N_2 abundance, …

Digital watch collects skin temperature, heart beat, … etc



Digital camera collects video information



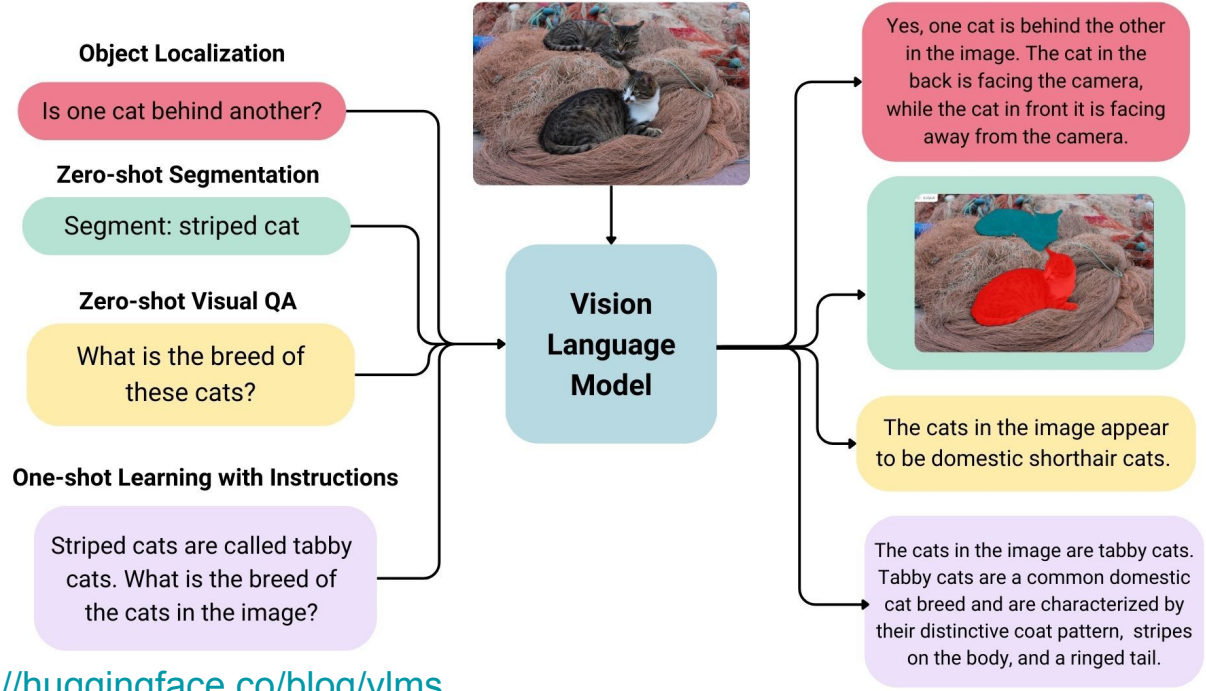
Spatial Intelligence

- Models the world and reason about objects, places and events in 3D space and time

Image credit: Google Image

Vision Language Models (VLMs)

- Multimodal (and generative) models that can learn from images and text
- Good zero-shot capabilities



Reference: https://huggingface.co/blog/vlms

Birds can also learn each other's languages

A Japanese doctor spent 20 years burrowing into the forest, intent on unlocking the secrets of the great tit. His discovery challenged human understanding: bird calls reveal a hidden vocabulary and grammar! A single sound can cause an entire flock of birds to misinterpret danger, while a change in order can completely alter the meaning. Is this just coincidence, or a true language? Watch this video to discover the truth about bird language that has rocked the scientific world.

研究 20 年，他聽懂了鳥語！/ 鳥也有文法錯誤？順序一改就聽不懂！/ 鳥類也能互相學語言！？| 青茶說

青茶說

462K views · 4 days ago

Reference: https://www.youtube.com/watch?v=Jb6Z0UvmP-Y

Final Messages

1. As an engineer, learn to embrace and master various generative AI tools
2. **Data-driven approaches** are the foundations of generative AI tools
3. If you pursue the career related to generative AI, do think about:
   - **A product, an embedded system** over an application
   - **A specialized agent** over a general agent

I wish you enjoy the journey of Data Science